Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses $\frac{1}{2}\int r^2\,d\theta$ or $\int_0^{\pi} r^2\,d\theta$ OE	AO1.1a	M1	$\frac{1}{2}\int(4+2\cos\theta)^2\,d\theta$
	Rewrites $\cos^2\theta$ in terms of $\cos 2\theta$	AO1.1a	M1	$\frac{1}{2}\int_{-\pi}^{\pi}(16+16\cos\theta+4\cos^2\theta)\,d\theta$
	Correctly integrates *their* expression, ft wrong non-zero coefficients.	AO1.1b	A1F	$\int_{-\pi}^{\pi}(8+8\cos\theta+(1+\cos 2\theta))\,d\theta$
	Obtains required answer from fully correct mathematical argument	AO2.1	R1	$\left[8\theta+8\sin\theta+\theta+\frac{1}{2}\sin 2\theta\right]_{-\pi}^{\pi}$ $=\left[9\theta+8\sin\theta+\frac{1}{2}\sin 2\theta\right]_{-\pi}^{\pi}$ $= (9\pi + 0 + 0) - (-9\pi + 0 + 0)$ $= 18\pi$
(b)	Selects appropriate method to determine polar equation by equating OA and OB to find θ	AO3.1a	M1	Let A(r_1, θ_1) and B(r_2, θ_2) OA = OB $\Rightarrow r_1 = r_2$ $\Rightarrow 4 + 2\cos\theta_1 = 4 + 2\cos\theta_2$
	Uses the above to find two values of θ and hence deduce the lengths of OA and OB Award this mark for correct deduction using 'their' values of θ	AO2.2a	R1	$\Rightarrow \theta_1 = -\theta_2$ Angle AOB = $\frac{\pi}{3}$ $\Rightarrow \theta_1 - \theta_2 = \frac{\pi}{3}$
	Uses the correct polar equation for a perpendicular line $r = d\sec\theta$	AO3.1a	M1	$\Rightarrow \theta_1 = \frac{\pi}{6}$ and $\theta_2 = -\frac{\pi}{6}$ OA = OB = $4 + \sqrt{3}$
	Obtains a correct equation for AB (including correct specified range) CAO	AO1.1b	A1	AB is perpendicular to the initial line Polar equation of AB is $r\cos\theta = (4+\sqrt{3})\frac{\sqrt{3}}{2}$ for

			$-\frac{\pi}{6} \le \theta \le \frac{\pi}{6}$
			Total 8 marks

Q2.

(a) (i) $r = \sin\frac{2\pi}{3}\sqrt{(2+\frac{1}{2}\cos\frac{\pi}{3})} = \frac{\sqrt{3}}{2}\times\sqrt{\frac{9}{4}} = \frac{3\sqrt{3}}{4}$

M1; A1

2

(ii) $x = ON = (3\sqrt{3})/8$

Polar eqn of PN is $r \cos \theta = ON$

M1

$r = \frac{3\sqrt{3}}{8}\sec\theta$

AG Be convinced

A1

2

(iii) Area Δ $ONP = 0.5 \times r_N \times r_P \times \sin(\pi/3)$

OE With correct or ft from (a)(i) (ii), values for r_P and r_N.

M1

$= \frac{1}{2}\times\frac{3\sqrt{3}}{8}\times\frac{3\sqrt{3}}{4}\times\frac{\sqrt{3}}{2} = \frac{27\sqrt{3}}{128}$

Be convinced

A1

2

(b) (i) $\int \sin^n \theta \cos \theta \, d\theta = \int u^n du$

PI

M1

$= \frac{\sin^{n+1}\theta}{n+1}$ $(+c)$

A1

2

(ii) Area of shaded region bounded by line OP and arc

$OP = \frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin^2 2\theta(2 + \frac{1}{2}\cos\theta)d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$

M1

Correct limits

B1

$$\frac{1}{2}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (1-\cos 4\theta)\,d\theta + \frac{1}{4}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 4\sin^2\theta\cos^2\theta\cos\theta\,d\theta$$

$2\sin^2 2\theta = \pm 1 \pm \cos 4\theta$

M1

$\sin^2 2\theta\cos\theta = 4\sin^2\theta\cos^2\theta\cos\theta$

B1

Correct integration of $0.5(1 - \cos 4\theta)$

A1

$$= \left[\frac{\theta}{2} - \frac{\sin 4\theta}{8}\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\sin^2\theta - \sin^4\theta)\cos\theta\,d\theta$$

Writing 2nd integrand in a suitable form to be able to use (b)(i) OE PI

m1

$$= \left[\frac{\theta}{2} - \frac{\sin 4\theta}{8} + \frac{\sin^3\theta}{3} - \frac{\sin^5\theta}{5}\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$$

Last two terms OE

A1

$$= \frac{\pi}{12} - \frac{21\sqrt{3}}{160} + \frac{2}{15}$$

CSO

A1

8

[16]

Q3.

(a) At intersections of $r = 2$ and $r = 3 + 2\sin\theta$

$2 = 3 + 2\sin\theta$

Elimation of r

M1

$\sin\theta = -\frac{1}{2}, \Rightarrow \theta = \frac{7}{6}\pi, \theta = \frac{11}{6}\pi$

Any one correct solution of $\sin\theta = -\frac{1}{2}$

A1

$(P=)\ \left(2, \frac{7\pi}{6}\right),\quad (Q=)\ \left(2, \frac{11\pi}{6}\right)$

$\left(2, \frac{7\pi}{6}\right)$ and $\left(2, \frac{11\pi}{6}\right)$

A1

3

(b) (i) Angle between OA and initial line $= \frac{\pi}{6}$

If not correct, ft on $\theta_P - \pi$

B1F

When $\theta = \frac{\pi}{6}$, $r = 3 + 2\sin\frac{\pi}{6} = 4$

$A\left(4, \frac{\pi}{6}\right)$

B1

2

(ii) $OA = 4$, $OQ = 2$

Angle $AOQ = \pi - (\theta_Q - \theta_P) = \frac{\pi}{3}$

If not correct, ft on $\pi - (\theta_Q - \theta_P)$

OE eg Cartesian coords of A *and* Q *both attempted and at least one correct ft.*

B1F

$AQ^2 = 4^2 + 2^2 - 2(4)(2)\cos AOQ\ (=12)$

Valid method to find AQ *(or* AQ^2*). Ft on c's* r_A *for* OA

M1

$AQ = \sqrt{12}$

ACF but must be exact surd form.

A1

3

(iii) Since $4^2 = 2^2 + (\sqrt{12})^2$ so 90°

Justifying why (angle OQA*=) 90° OE*

E1

angle OQA = 90° ⇒ AQ is a tangent

Must have convincingly shown that OQA = 90°

E1

2

(c) Area of minor sector OPQ of circle = $\frac{1}{2}(2)^2[\theta_Q - \theta_P]$

$\frac{1}{2}(2)^2[\theta_Q - \theta_P]$

M1

$= \frac{4\pi}{3}$

PI by combined $-\frac{7\pi}{3}$ OE term later.

A1

Area of minor region OPQ of curve = $\frac{1}{2}\int_{\frac{7\pi}{6}}^{\frac{11\pi}{6}}$ $(4\sin^2\theta + 12\sin\theta + 9)\,d\theta$

Use of $\frac{1}{2}\int$ $r^2\,d\theta$ or use of $\int_{\theta_P}^{3\pi/2} r^2$ $d\theta$ OE

M1

$r^2 = 4\sin^2\theta + 12\sin\theta + 9$

B1

$= \frac{1}{2}\int(2 - 2\cos 2\theta + 12\sin\theta + 9)d\theta$

Use of $\cos 2\theta = \pm 1 \pm 2\sin^2\theta$ with $k\int r^2$ $(d\theta)$

M1

$= \frac{1}{2}$ $[2\theta - \sin 2\theta - 12\cos\theta + 9\theta]$ =

Ft wrong non zero coefficients, ie for correct integration of $a + b\cos 2\theta + c\sin\theta$

A1F

$$\left[\frac{121\pi}{12} + \frac{\sqrt{3}}{4} - \frac{6\sqrt{3}}{2}\right] - \left[\frac{77\pi}{12} - \frac{\sqrt{3}}{4} + \frac{6\sqrt{3}}{2}\right]$$

$\{= \frac{11\pi}{3} - \frac{11\sqrt{3}}{2}\}$

OE eg $\left[\frac{33\pi}{2}\right] - \left[\frac{77\pi}{6} - \frac{\sqrt{3}}{2} + 6\sqrt{3}\right]$

$$eg\left[\frac{121\pi}{6}+\frac{\sqrt{3}}{2}-6\sqrt{3}\right]-\left[\frac{33\pi}{2}\right]$$

A1

Area of shaded region = $\frac{4\pi}{3}-\{\frac{11\pi}{3}-\frac{11\sqrt{3}}{2}\}$

$\frac{1}{2}(2)^2\left[\theta_Q-\theta_P\right]-\frac{1}{2}\int_{\theta_P}^{\theta_Q}$ $(3 + \sin\theta)^2\, d\theta$

M1

$=\frac{11}{2}\sqrt{3}-\frac{7}{3}\pi=\frac{1}{6}\left(33\sqrt{3}-14\pi\right)$

CSO $\frac{1}{6}\left(33\sqrt{3}-14\pi\right)$. ($m = 33$, $n = -14$)

A1

9

[19]

Q4.

(a) Area = $\frac{1}{2}\int(3 + 2\cos\theta)^2\, d\theta$

Use of $\frac{1}{2}\int r^2\, d\theta$ or $\int_0^{\pi} r^2\, d\theta$

M1

$=\frac{1}{2}\int_0^{2\pi}(9 + 12\cos\theta + 4\cos^2\theta)\, d\theta$

Correct expn of $[3 + 2\cos\theta]^2$

B1

Correct limits

B1

$=\int_0^{2\pi}(4.5 + 6\cos\theta + (1 + \cos 2\theta))\, d\theta$

Attempt to write $\cos^2\theta$ in terms of $\cos 2\theta$

M1

$=\left[4.5\theta + 6\sin\theta + \theta + \frac{1}{2}\sin 2\theta\right]_0^{2\pi}$

Correct integration ft wrong coefficients

A1F

$= 11\pi$

CSO

A1

6

(b) (i) $x^2 + y^2 - 8x + 16 = 16$

Use of ***any two*** *of* $x = r\cos\theta$, $y = r\sin\theta$, $x^2 + y^2 = r^2$

M1

$r^2 - 8r\cos\theta + 16 = 16 \Rightarrow r = 8\cos\theta$

A1

At intersection, $8\cos\theta = 3 + 2\cos\theta \Rightarrow \cos\theta = \frac{3}{6}$

Equating rs or equating cosθs with a further step to solve eqn.
(OE eg $4r = 12 + r \Rightarrow 4r - r = 12$)

M1

Points $\left(4, \frac{\pi}{3}\right)$ and $\left(4, \frac{5\pi}{3}\right)$

OE

A1

$AB = 2 \times \left(4\sin\frac{\pi}{3}\right)$

Valid method to find AB, ft c's r and θ values

M1

$= 4\sqrt{3}$

OE surd

A1

Alternative:

Writing $r = 3 + 2\cos\theta$ in cartesian form

(M1A1)

Finding cartesian coordinates of points A and B ie $(2, \pm 2\sqrt{2})$

(M1A1)

Finding length AB

(M1A1)

6

(ii) Let M = centre of circle then $\angle ABM = \frac{2\pi}{3}$

Accept equiv eg $\angle AMO = \frac{\pi}{3}$

B1

Length of arc AOB of circle = $4 \times \frac{2\pi}{3}$

Use of arc = 4 × ($\angle AMB$ *in rads)*

Perimeter of segment $AOB = \frac{8\pi}{3} + 4\sqrt{3}$

A1

3

[15]

Q5.

Area = $\frac{1}{2}\int \left(2\sqrt{1+\tan\theta}\right)^2$ $(d\theta)$

Use of $\frac{1}{2}\int r^2$ $(d\theta)$

M1

$= \frac{1}{2}\int_{-\frac{\pi}{4}}^{0} 4(1 + \tan\theta)\,d\theta$

Correct limits. If any contradiction use the limits at the substitution stage

B1

$= 2\left[\theta + \ln\sec\theta\right]_{-\frac{\pi}{4}}^{0}$

$\int k(1 + \tan\theta)$ $(d\theta)$ = $k(\theta + \ln\sec\theta)$

ACF ft on c's k

B1

$= 2\left\{0 - \left[-\frac{\pi}{4} + \ln\sec\left(-\frac{\pi}{4}\right)\right]\right\}$

$= 2\left(\frac{\pi}{4} - \ln\sqrt{2}\right) = \frac{\pi}{2} - 2\ln\sqrt{2} = \frac{\pi}{2} - \ln 2$

CSO AG

A1

[4]

Q6.

$$\text{Area} = \frac{1}{2}\int\left(2\sin 2\theta\sqrt{\cos\theta}\right)^2 \mathrm{d}\theta$$

Use of $\frac{1}{2}\int r^2 \mathrm{d}\theta$

M1

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}}\left(4\cos\theta\sin^2 \theta\right)\mathrm{d}\theta$$

$r^2 = 4\cos\theta\sin^2 2\theta$ or better

B1

Correct limits

B1

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}}\left(16\sin^2\theta\cos^3\theta\right)\mathrm{d}\theta$$

$\sin^2 2\theta = k\sin^2\theta\cos^2\theta\ (k > 0)$

M1

$$= \int_0^{\frac{\pi}{2}}\left(8\sin^2\theta\left(1-\sin^2\theta\right)\right)\mathrm{d}\sin\theta$$

Substitution or another valid method to integrate $\sin^2\theta\cos^3\theta$

m1

$$= \left[\frac{8\sin^3\theta}{3} - \frac{8\sin^5\theta}{5}\right]_0^{\frac{\pi}{2}}$$

Correct integration of $p\sin^2\theta\cos^3\theta$

A1F

$$= \left(\frac{8}{3} - \frac{8}{5}\right) - 0 = \frac{16}{5}$$

CSO AG

A1

7

Alternatives for the last four marks

$$\text{Area} = \int_0^{\frac{\pi}{2}} (\cos\theta - \cos 4\theta \cos\theta) \mathrm{d}\theta$$

2 cosθ sin² 2θ = λ cosθ + μ cos 4θ cosθ
(λ, $\mu \neq 0$)

(M1)

$$\int (\cos 4\theta \cos\theta) \mathrm{d}\theta$$

Integration by parts twice or use of

$$\cos 4\theta \cos\theta = \frac{1}{2}(\cos 5\theta + \cos 3\theta)$$

(m1)

$$= -\frac{1}{15}(\cos 4\theta \sin\theta - 4 \sin 4\theta \cos\theta)$$

Correct integration of p cos 4θ cosθ

$$[eg\ p\left[\frac{1}{10}\sin 5\theta + \frac{1}{6}\sin 3\theta\right]]$$

(A1F)

$$\text{Area} = (1-0) + \frac{1}{15}[(1-0)-(0)] = \frac{16}{15}$$

CSO AG

$$\left\{1 - \frac{1}{10} + \frac{1}{6} = \frac{16}{15}\right\}$$

(A1)

(7)

[7]

Q7.

(a) $r = 2\sin\theta \Rightarrow r^2 = 2r\sin\theta$

M1

$x^2 + y^2 = 2y$

OE (A1) either for $r^2 = x^2 + y^2$ or for $r\sin\theta = y$

***SC** If M0 give B1 for $r^2 = x^2 + y^2$ or for $r\sin\theta = y$ used*

A2,1

3

(b) (i) 2 sin θ = tan θ

Equating rs

M1

$2\sin\theta\cos\theta = \sin\theta$

$\sin\theta\,(2\cos\theta - 1) = 0$

Both solutions have to be considered if not in factorised form

m1

$\sin\theta = 0 \Rightarrow \theta = 0; \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$

Alternative: $\sin 2\theta = \sin\theta \Rightarrow \theta = 0, \frac{\pi}{3}$

$\theta = 0 \Rightarrow r = 0$ ie pole O (0,0)

Indep. Can just verify using both eqns + statement.

B1

$\theta = \frac{\pi}{3} \Rightarrow r = \sqrt{3}\ \left(P\left(\sqrt{3}, \frac{\pi}{3}\right)\right)$

CSO

A1

4

(ii) At A, $\theta = \frac{\pi}{4}, r = 2\sin\frac{\pi}{4} = \sqrt{2}$

At B, $\theta = \frac{\pi}{4}, r = \tan\frac{\pi}{4} = 1$

Substitute $\theta = \frac{\pi}{4}$ *into the equations of both curves.*

M1

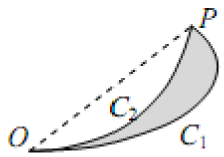
Since $\sqrt{2}$ >1, A is further away (from the pole than B.)

CSO

E1

2

(iii)



Area bounded by line *OP* and curve C_1

$= \frac{1}{2}\int_0^{\frac{\pi}{3}} 4\sin^2\theta \, d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$ *; ignore limits here*

M1

$= \int (1 - \cos 2\theta) \, d\theta$

Attempt to write $\sin^2\theta$ *in terms of* $\cos 2\theta$ *only*

m1

$= \left[\theta - \frac{1}{2}\sin 2\theta\right]$

Ignore limits here

A1

$= \left(\frac{\pi}{3} - \frac{1}{2} \times \frac{\sqrt{3}}{2}\right) - 0 = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$

PI

A1

Area bounded by line *OP* and curve C_2

$= \frac{1}{2}\int_0^{\frac{\pi}{3}} \tan^2\theta \, d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$ *; ignore limits here*

M1

$= \frac{1}{2}\int (\sec^2\theta - 1) \, d\theta$

Using $\tan^2\theta = \pm \sec^2\theta \pm 1$ *PI*

m1

$= \frac{1}{2}[\tan\theta - \theta]$

Ignore limits here

A1

$= \frac{1}{2}\left(\sqrt{3} - \frac{\pi}{3}\right) - 0 = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$

PI

A1

Required area $= \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4}\right) - \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right)$

Can award earlier eg if we see

$\frac{1}{2}\int_0^{\frac{\pi}{3}} 4\sin^2\theta\, d\theta - \frac{1}{2}\int_0^{\frac{\pi}{3}} \tan^2\theta\, d\theta$

M1

$= \frac{1}{2}\pi - \frac{3}{4}\sqrt{3} \quad \left(a = \frac{1}{2},\ b = -\frac{3}{4}\right)$

CSO

A1

10

[19]

Q8.

(a) $4 \sin \theta (1 - \sin \theta) = 1$

Elimination of r or θ $\{r = 4[1 - (1/r)]\}$

M1

$4 \sin^2 \theta - 4 \sin \theta + 1 = 0$

$\{r^2 - 4r + 4 = 0\}$

A1

$(2 \sin \theta - 1)^2 = 0 \Rightarrow \sin \theta. =0.5$

Valid method to solve quadratic eqn. PI

$\{(r - 2)^2 = 0 \Rightarrow r = 2\}$

m1

$\theta = \frac{\pi}{6},\ \theta = \frac{5\pi}{6}, r = 2$

$[P\left(2, \frac{\pi}{6}\right) \quad Q\left(2, \frac{5\pi}{6}\right)]$

A1 for any two of the three.

A2,1

SC: Verification of $P\left(2, \frac{\pi}{6}\right)$ *scores max*

of B1 & a further B1 if $Q\left(2, \frac{5\pi}{6}\right)$ *stated*

5

(b) Area triangle $OPQ = \frac{1}{2} \times 2 \times r_Q \times \sin POQ$

Any valid method to correct (ft eg on r_Q) expression with just one remaining unknown

M1

Angle $POQ = \frac{5\pi}{6} - \frac{\pi}{6} \quad \left(= \frac{2\pi}{3}\right)$

Valid method to find remaining unknown either relevant angle or relevant side

m1

Area triangle $OPQ = 2\sin\frac{2\pi}{3} = \sqrt{3}$

A1

Unshaded area bounded by line OP and arc OP =

$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} [4(1 - \sin\theta)]^2 \, d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$ *for relevant area(s) (condone missing/wrong limits)*

M1

$= 8\int (1 - 2\sin\theta + \sin^2\theta) d\theta$

Correct expn of $(1 - \sin\theta)^2$

B1

$= 8\int \left(1 - 2\sin\theta + \frac{1 - \cos 2\theta}{2}\right) d\theta$

Attempt to write $\sin^2\theta$ *in terms of* $\cos 2\theta$

M1

$= 8\left[\theta + 2\sin\theta + \frac{\theta}{2} - \frac{\sin 2\theta}{4}\right] (+c)$

Correct integration ft wrong coeffs

A1F

$$8\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}(1-\sin\theta)^2\ d\theta = 8\times\left[\frac{3\theta}{2}+2\cos\theta-\frac{\sin 2\theta}{4}\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= 8\times\{\frac{3\pi}{4}-\left(\frac{3\pi}{12}+2\cos\frac{\pi}{6}-\frac{1}{4}\sin\frac{2\pi}{6}\right)\}$$

$F\left(\frac{\pi}{2}\right)-F\left(\frac{\pi}{6}\right)$ *OE for relevant area(s)*

m1

$$= 8\times\left(\frac{\pi}{2}-\sqrt{3}+\frac{\sqrt{3}}{8}\right)\quad \{= 4\pi-7\sqrt{3})$$

ft one slip; accept terms in π and $\sqrt{3}$ left unsimplified

A1F

Shaded area = Area of triangle OPQ –

$$2\times\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}[4(1-\sin\theta)]^2\ d\theta$$

OE

M1

Shaded area = $\sqrt{3}-16\left(\frac{\pi}{2}-\sqrt{3}+\frac{\sqrt{3}}{8}\right)=15\sqrt{3}-8\pi$

CSO *Accept m = 15, n = –8*

A1 **11**

[16]

Q9.

(a) (i) $x^2+y^2=r^2$, $x=r\cos\theta$, $y=r\sin\theta$

B1 for one stated or used

B2,1,0

$r^2 = 2r(\cos\theta-\sin\theta)$

M1

$x^2+y^2=2(x-y)$

ACF

A1 **4**

(ii) $(x-1)^2 + (y+1)^2 = 2$

M1A1F

Centre (1, –1); radius $\sqrt{2}$

A1F

3

(b) (i) $\text{Area} = \frac{1}{2}\int (4+\sin\theta)^2 \, d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$.

M1

$= \frac{1}{2}\int_0^{2\pi} (16 + 8\sin\theta + \sin^2\theta) \, d\theta$

Correct expn of $[4+\sin\theta]^2$

B1

Correct limits

B1

$= \int_0^{2\pi} (8 + 4\sin\theta + 0.25(1-\cos 2\theta)) \, d\theta$

Attempt to write $\sin^2\theta$ *in terms of* $\cos 2\theta$

M1

$= \left[8\theta - 4\cos\theta + \frac{1}{4}\theta - \frac{1}{8}\sin 2\theta\right]_0^{2\pi}$

Correct integration ft wrong coefficients

A1F

$= 16.5\pi$

CSO

A1

6

(ii) For the curves to intersect, the eqn
$2(\cos\theta - \sin\theta) = 4 + \sin\theta$
must have a solution.

Equating rs and simplifying to a suitable form

M1

$2\cos\theta - 3\sin\theta = 4$

$R\cos(\theta + \alpha) = 4$.

OE. Forming a relevant eqn from which valid explanation can be stated directly

M1

where $R = \sqrt{2^2 + 3^2}$ and cos $\alpha = \frac{2}{R}$

OE. Correct relevant equation

A1

$\cos(\theta + \alpha) = \frac{4}{\sqrt{13}} > 1$. Since must have $-1 \le \cos X \le 1$ there are no solutions of the equation $2(\cos\theta - \sin\theta) = 4 + \sin\theta$ so the two curves do not intersect.

Accept other valid explanations.

E1

4

(iii) Required area = answer (b)(i) – π(radius of C_1)^2

M1

$= 16.5\pi - 2\pi = 14.5\pi$

Ft on (a)(ii) and (b)(i)

A1F

2

[19]

Q10.

Area = $\frac{1}{2}\int_0^{\pi}(2+\cos\theta)^2\sin\theta\,d\theta$

use of $\frac{1}{2}\int r^2\,d\theta$

M1

Correct limits

B1

$= \frac{1}{2}\left[-\frac{1}{3}(2+\cos\pi)^3\right]_0^{\pi}$

Valid method to reach
$k(2 + \cos\theta)^3$ or $a\cos\theta + b\cos 2\theta + c\cos^3\theta$ OE
{SC: M1 if expands then integrates to get
either $a\cos\theta + b\cos 2\theta$ OE or

c $\cos^3\theta$ *OE in a valid way}*

M2

OE eg $-4\cos\theta - \cos 2\theta - \frac{1}{3}\cos^3\theta$

A1

$$= \frac{1}{2}\left\{-\frac{1}{3} + \frac{1}{3}\times 3^3\right\} = \frac{13}{3}$$

CSO

A1

[6]

Q11.

(a)

$$\text{Area} = \frac{1}{2}\int\left(1 + 6e^{-\frac{\theta}{\pi}}\right) d\theta$$

Use of $\frac{1}{2}\int r^2\, d\theta$

M1

$$= \frac{1}{2}\int_0^{2\pi}\left(1 + 12e^{-\frac{\theta}{\pi}} + 36e^{-\frac{2\theta}{\pi}}\right) d\theta$$

Correct expansion of $\left(1 + 6e^{-\frac{\theta}{\pi}}\right)^2$

B1

Correct limits

B1

$$= \frac{1}{2}\left[\theta - 12\pi e^{-\frac{\theta}{\pi}} - 18\pi e^{-\frac{2\theta}{\pi}}\right]_0^{2\pi}$$

Correct integration of at least two of the three terms 1, $pe^{-\frac{\theta}{\pi}}$, $qe^{-\frac{2\theta}{\pi}}$

m1

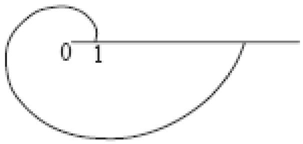
$= \pi(16 - 6e^{-2} - 9e^{-4})$

ACF

A1

5

(b)



Going the correct way round the pole

B1

Increasing in distance from the pole

B1

End-points (1, 0) and (e^2, 2π)

Correct end-points
B1 for each pair or for 1 and e^2 shown on graph in correct positions

B2,1,0

4

(c) $e^{\frac{\theta}{\pi}} = 1 + 6e^{-\frac{\theta}{\pi}}$

Elimination of r or θ $[r = 1 + \frac{6}{r}]$

M1

$\left(e^{\frac{\theta}{\pi}}\right)^2 - e^{\frac{\theta}{\pi}} - 6 = 0$

Forming quadratic in $e^{\frac{\theta}{\pi}}$ or in $e^{-\frac{\theta}{\pi}}$ or in r. $[r^2 - r - 6 = 0]$

m1

$\left(e^{\frac{\theta}{\pi}} - 3\right)\left(e^{\frac{\theta}{\pi}} + 2\right) = 0$

OE

m1

$e^{\frac{\theta}{\pi}} > 0$ so $e^{\frac{\theta}{\pi}} = 3$

Rejection of negative 'solution' PI
[r = 3]

E1

Polar coordinates of P are (3, π ln 3)

A1

5

[14]

Q12.

(a) Area = $\frac{1}{2}\int(1+\tan\theta)^2\,\mathrm{d}\theta$

Use of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$

M1

$\ldots = \frac{1}{2}\int(1+2\tan\theta+\tan^2\theta)\,\mathrm{d}\theta$

Correct expansion of (1 + tan θ)²

B1

$= \frac{1}{2}\int(\sec^2\theta+2\tan\theta)\,\mathrm{d}\theta$

1 + tan² θ = sec² θ used

M1

$= \frac{1}{2}\left[\tan\theta+2\ln(\sec\theta)\right]_0^{\frac{\pi}{3}}$

Integrating psec² θ correctly

A1ft

Integrating qtan θ correctly

B1ft

$= \frac{1}{2}[(\sqrt{3}+2\ln 2)-0] = \frac{\sqrt{3}}{2}+\ln 2$

Completion. AG CSO be convinced

A1

6

(b) $OP = 1$; $OQ = 1 + \tan\frac{\pi}{3}$

Both needed. Accept 2.73 for OQ

B1

Shaded area = 'answer (a)' $-\frac{1}{2}OP\times OQ\times\sin\left(\frac{\pi}{3}\right)$

M1

$= \frac{\sqrt{3}}{2}+\ln 2-\frac{\sqrt{3}}{4}(1+\sqrt{3})$

ACF. Condone 0.376… if exact 'value' for area of triangle seen

A1

$= \frac{\sqrt{3}}{4} + \ln 2 - \frac{3}{4}$

3

[9]

Q13.

(a) $\theta = 0$, $r = 5 + 2\cos 0 = 7$ {A lies on C}

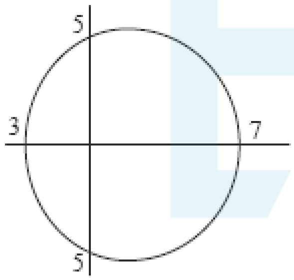
B1

$\theta = \pi$, $r = 5 + 2\cos\pi = 3$ {B lies on C}

B1

2

(b)



Closed single loop curve, with (indication of) symmetry

B1

Critical values, 3, 5, 7 indicated

B1

2

(c) $\text{Area} = \frac{1}{2}\int (5 + 2\cos\theta)^2 \, d\theta$

Use of $\frac{1}{2}\int r^2 \, d\theta$

M1

$= \frac{1}{2}\int_{-\pi}^{\pi} (25 + 20\cos\theta + 4\cos^2\theta) \, d\theta$

OE for correct expansion of $(5 + 2\cos\theta)^2$

B1

For correct limits

B1

$$= \frac{1}{2}\int_{-\pi}^{\pi}(25 + 20\cos\theta + 2(\cos 2\theta + 1))\,d\theta$$

Attempt to write cos² θ in terms of cos 2θ

M1

$$= \frac{1}{2}[27\theta + 20\sin\theta + \sin 2\theta]_{-\pi}^{\pi}$$

Correct integration ft wrong non-zero coefficients in a + b cos θ + c cos 2θ

A1F

= 27π

CSO

A1 **6**

(d) Triangle OBQ with
OB = 3 and angle BOQ = α

PI

B1

OQ = 5 + 2 cos(–π + α)

OE

M1

Area of triangle OQB = $\frac{1}{2}OB \times OQ \sin\alpha$

Dep. on correct method to find OQ

m1

$$= \frac{3}{2}(5 - 2\cos\alpha)\sin\alpha$$

CSO

A1 **4**

[14]

Q14.

(a) Area = $\frac{1}{2}\int (6 + \cos\theta)^2\,d\theta$

use of $\frac{1}{2}\int r^2 \, d\theta$

M1

$$= \frac{1}{2}\left(\int_{-\pi}^{\pi} 36 + 48\cos\theta + 16\cos^2\theta\right) d\theta$$

for correct expansion of [6 + 4cosθ)]^2

B1

for limits

B1

$$= \left(\int_{-\pi}^{\pi} 18 + 24\cos\theta + 4(\cos 2\theta + 1)\right) d\theta$$

Attempt to write cos^2 θ in terms of cos 2θ

M1

$$= [22\theta + 24\sin\theta + 2\sin 2\theta]_{-\pi}^{\pi}$$

correct integration ft wrong coefficients

A1F

$= 44\pi$

CSO

A1

6

(b) At P, $r = 4$; At Q, $r = 2$;

PI

B1

P {x = } $r\cos\theta$ $= 4\cos\frac{2\pi}{3} = -2$

Attempt to use r cos θ

M1

$Q\{x = \}\, r\cos\theta = 2\cos\pi = -2$

Both

A1

Since P and Q have same 'x', PQ is vertical so QP is parallel to the vertical line $\theta = \frac{\pi}{2}$

E1

4

(c) (i) $OP = 4$; $OS = 8$;

B1

Angle $POS = \frac{\pi}{3}$

or S (4, $4\sqrt{3}$) and P (–2, $2\sqrt{3}$)

B1

$PS^2 = 4^2 + 8^2 - 2 \times 4 \times 8 \times \cos\frac{\pi}{3}$ *oe*

Cosine rule used in triangle POS

OE $PS^2 = (4+2)^2 + (4\sqrt{3} - 2\sqrt{3})^2$

M1

$PS = \sqrt{48}\ \{= 4\sqrt{3}\}$

A1

4

(ii) Since $8^2 = 4^2 + (\sqrt{48})^2$,

Accept valid equivalents e.g.

$PR = 2PQ = 2(2\sqrt{3}) = PS$.

E1

1

$OS^2 = OP^2 + PS^2 \Rightarrow OPS$ is a right angle.

(Converse of Pythagoras Theorem)

$\angle SRP = \angle RSP = \angle RPO = \frac{\pi}{6}$

$\Rightarrow OPS$ is a right angle

[15]

Q15.

(a) $(\cos\theta + \sin\theta)^2 = \cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta = 1 + \sin 2\theta$

AG (be convinced)

B1

1

(b) $(x^2 + y^2)^3 = (x + y)^4$

$(r^2)^3 = (r\cos\theta + r\sin\theta)^4$

[M1 for one of $x^2 + y^2 = r^2$ OE,

$x = r\cos\theta$, $y = r\sin\theta$ used]

M2,1,0

$r^6 = r^4(\cos\theta + \sin\theta)^4$

$r^6 = r^4(1 + \sin 2\theta)^2$

Uses (a) OE at any stage

M1

$r^2 = (1 + \sin 2\theta)^2$

$\Rightarrow r = (1 + \sin 2\theta)\ \{r \geq 0\}$

CSO; AG

A1

4

(c) (i) $r = 0 \Rightarrow \sin 2\theta = -1$

$2\theta = \sin^{-1}(-1); = -\frac{\pi}{2}, \frac{3\pi}{2}$

M1

$\theta = -\frac{\pi}{4}; \frac{3\pi}{4}$

A1 for either

A1A1ft

3

(ii) Area $= \frac{1}{2}\int (1 + \sin 2\theta)^2\, d\theta$

Use of $\frac{1}{2}\int r^2 d\theta$

M1

$= \frac{1}{2}\int \left(1 + 2\sin 2\theta + \sin^2 2\theta\right) d\theta$

Correct expansion of $(1 + \sin 2\theta)^2$

B1

$= \frac{1}{2}\int \left(1 + 2\sin 2\theta + \frac{1}{2}(1 - \cos 4\theta)\right) d\theta$

Attempt to write $\sin^2 2\theta$ in terms of $\cos 4\theta$

M1

$$= \left[\frac{3}{4}\theta - \frac{1}{2}\cos 2\theta - \frac{1}{16}\sin 4\theta\right]$$

Correct integration ft wrong coefficients only

A1ft

$$= \left[\frac{3}{4}\theta - \frac{1}{2}\cos 2\theta - \frac{1}{16}\sin 4\theta\right]_{-\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \left(\frac{9\pi}{16}\right) - \left(-\frac{3\pi}{16}\right)$$

Using c's values from (c)(i) as limits or the correct limits

m1

$$= \frac{3\pi}{4}$$

CSO

A1 6

[14]

Q16.

(a) $x^2 + y^2 - 12y + 36 = 36$

Use of $y = r\sin\theta$ ($x = r\cos\theta$ PI)

M1

Use of $x^2 + y^2 = r^2$

M1

$r^2 - 12r\sin\theta + 36 = 36$

m1

$\Rightarrow r = 12\sin\theta$

CSO AG

A1 4

(b) Area = $\frac{1}{2}\int (2\sin\theta + 5)^2\, d\theta$

Use of $\frac{1}{2}\int r^2\, d\theta$.

M1

$.. = \frac{1}{2}\int_0^{2\pi}(4\sin^2\theta + 20\sin\theta + 25)d\theta$

Correct expn. of (2sin(θ + 5)^2

B1

Correct limits

B1

$= \frac{1}{2}\int_0^{2\pi} 2(1-\cos 2\theta) + 20\sin\theta + 25)$ dθ

Attempt to write sin$^2\theta$ in terms of cos 2θ.

M1

$= \frac{1}{2}\left[27\theta - \sin 2\theta - 20\cos\theta\right]_0^{2\pi}$

Correct integration ft wrong coeffs

A1√

$= 27\pi$.

CSO

A1

6

(c) At intersection 12 sin θ = 2 sin θ + 5

OE eg r = 6(r − 5)

M1

$\Rightarrow \sin\theta = \frac{5}{10}$

OE eg r = 6

A1

Points $\left(6, \frac{\pi}{6}\right)$ and $\left(6, \frac{5\pi}{6}\right)$

OE

A1

$OPMQ$ is a rhombus of side 6

Or two equilateral triangles of side 6

Area = 6 × 6 × sin $\frac{2\pi}{3}$ oe

Any valid complete method to find the

area (or half area) of quadrilateral.

M1
A1

= $18\sqrt{3}$

Accept unsimplified surd

Al

6

[16]

Q17.

(a) Area = $\frac{1}{2}\int 36(1-\cos\theta)^2\,\mathrm{d}\theta$

use of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$

M1

…= $\frac{1}{2}\int_0^{2\pi} 36(1-2\cos\theta+\cos^2\theta)\,\mathrm{d}\theta$

B1

for correct explanation of $[6(1-\cos\theta)]^2$

for correct limits

B1

= $9\int_0^{2\pi} 2-4\cos\theta+(\cos 2\theta+1)\,\mathrm{d}\theta$

Attempt to write $\cos^2\theta$ *in terms of cos* 2θ*.*

M1

= $\left[27\theta-36\sin\theta+\frac{9}{2}\sin 2\theta\right]_0^{2\pi}$

Correct integration; only ft if integrating
$a + b\cos\theta + c\cos 2\theta$ with non-zero a, b, c.

A1ft

= 54π

CSO

A1

6

(b) (i) $x^2 + y^2 = 9 \Rightarrow r^2 = 9$

PI

B1

A & B: $3 = 6 - 6\cos\theta \Rightarrow \cos\theta = \frac{1}{2}$

M1

Pts of intersection $\left(3, \frac{\pi}{3}\right)$; $\left(3, \frac{5\pi}{3}\right)$

OE (accept 'different' values of θ not in the given interval)

A1A1ft

4

(ii) Length AB = $2 \times r \sin\theta$

M1

………… = $2 \times 3 \times \frac{\sqrt{3}}{2} = 3\sqrt{3}$

OE exact surd form

A1

2

[12]

Q2.

As expected, this final question proved to be the most demanding one for the students with very few presenting a full correct solution. Part (a)(i) was almost always answered correctly but part (a)(ii) proved to be more of a challenge with a significant number of non-attempts. Those students who found the length of ON, by using their answer to part (a)(i) and basic trigonometry for the right-angled triangle shown in the given diagram, generally stated the correct Cartesian equation of the line PN and gave its polar equation by using $x = r\cos\theta$ and rearranging to give the answer in the printed form. Students were much more successful in finding the area of the triangle in part (a)(iii). Although most students correctly used the lengths of the sides of triangle OPN to find the correct area it was also pleasing to see the alternative approach of integrating half the square of the expression for r in part (a)(ii) between the limits 0 and $\frac{\pi}{3}$. There were many correct answers for part (b)(i) although not all students gave their final answer in terms of θ. The final part of the question proved to be beyond many students although most did pick up partial credit. Deciphering what limits to use was an initial problem, with a significant number of students choosing 0 as one of the limits and then referring later to the 'bit of a' region between ON and the curve. It was also apparent that a significant number of students did not seem to appreciate the relevance of (b)(i). The most successful attempts considered the integration of $\sin^2 2\theta$ and the integration of $\frac{1}{4}\sin^2 2\theta\cos\theta$ separately, the first being found by use of the double angle identity to introduce $\cos 4\theta$ and the second being found by writing the integrand in a form which matched the integrand in part (b)(i) for $n = 2$ and $n = 4$.

Q3.

This question on polar coordinates produced some excellent solutions, particularly in the final part of the question. Most students scored at least two of the three marks in part (a), but a significant proportion gave a θ-coordinate that was outside the stated interval. There were many students who gave the correct polar coordinates of A in part (b)(i). In part (b)(ii) the most common correct methods for finding the length of AQ were either to use the cosine rule in triangle AOQ or use of the Cartesian coordinates of the points A and Q. A significant number of students failed to gain credit because they just assumed that angle AQO was a right angle and then used Pythagoras to find AQ, some then claiming in the next part that since Pythagoras was true the angle was 90° and so the line was a tangent. Those students who made better attempts at explaining why AQ was a tangent sometimes failed to score both marks because they either did not show sufficient detail or did not explicitly identify the relevant angle within their explanations. In part (c) many students showed that they knew the method for finding the area of the minor region OPQ of the curve. A significant number of students realised that the minor sector OPQ of the circle was required but many good attempts at combining these results were spoilt by an incorrect sign at some stage, normally in one of the surd forms for a trigonometric term involving $\frac{7}{6}\pi$ or $\frac{11}{6}\pi$.

Q4.

Most candidates had a good understanding of the method required to find the area of the region bounded by the curve. It was pleasing to see that the method for integrating $\cos^2\theta$ was well known, with many candidates scoring at least 4 of the 6 marks.

In part (b)(i) those candidates who started by finding the polar equation for the circle were generally more successful than those who started by finding the cartesian equation of the curve. Some candidates incorrectly stated the polar equation of the circle as $r = 4$, presumably just by consideration of the radius rather than any thought about the centre of the circle.

The final part of this final question proved to be very demanding for almost all candidates. Many answers displayed a lack of understanding of the term 'segment', with most solutions involving a perimeter of a region which consisted of the two radii and either an arc length (i.e. perimeter of a sector) or the length of AB (i.e. perimeter of a triangle). The few successful candidates noted that the circle passed through the pole, O, drew the circle and curve on the same diagram and found the size of angle AMB, in radians, with M being the centre of the circle. They then found the arc length using MPC2 work and added it to the length of AB found in part (b)(i).

Q5.

Most students were able to make good progress with this question which tested finding an area involving a curve whose polar equation was given. The most common error was the use of incorrect limits and this was frequently followed by further errors so as to match the printed answer. A number of students lost the final accuracy mark because their solution involved an incorrect line of working, the most common one being substitution of limits for θ into the mixed expression (θ In sec x) .

Q6.

Although most candidates scored marks for substituting a correct expression for r^2 into $A = \frac{1}{2}\int r^2\,d\theta$ and inserting the correct limits, less able candidates made little further progress. A significant proportion of other candidates went further by either using the identity $\sin 2\theta = 2\sin\theta\cos\theta$ or writing $\sin^2 2\theta$ in terms of $\cos 4\theta$. A surprisingly common error amongst those solutions which used the latter approach is illustrated by '$(1-\cos 4\theta)\cos\theta = \cos\theta - \cos^2 4\theta$'. Some excellent solutions were seen from the more able candidates, which involved a variety of approaches including integration of $8\sin^2\theta\cos^3\theta$ by use of the substitution $s = \sin\theta$, direct integration by recalling that the integral of $\sin^n\theta\cos\theta$ with respect to θ is $\frac{\sin^{n+1}\theta}{n+1}$ or use of the identity $2\cos 4\theta\cos\theta = \cos 5\theta + \cos 3\theta$. Those candidates who used integration by parts were generally less successful. Only a few obtained the correct answer by applying integration by parts twice.

Q7.

This final question tested polar coordinates and proved to be the most demanding question on the paper. In part (a), it was pleasing to see that the majority of candidates were able to use a correct method to find the cartesian equation. Those candidates who attempted to square both sides of the polar equation were generally less successful than those who started by multiplying both sides by r. In part (b)(i), a large number of candidates only considered the solution to $\cos\theta = 0.5$ when solving $2\sin\theta = \tan\theta$ and so failed to prove that the curves only met at the two points. Those candidates who attempted to verify that the curves intersected at the pole often substituted $\theta = 0$ rather than $r = 0$ into the two equations, but then often failed to give a final statement. In part (b)(ii), candidates generally substituted $\theta = \frac{\pi}{4}$ correctly into the two equations but a significant minority then did not give a full justification of whether A or B was further away from the pole. Although only the better candidates understood the relevance of part (b)(ii) in finding the required area in part (b)(iii), many candidates were able to make good progress in this part of the question. It was, however, disappointing to see some solutions that did not even use the correct formula for the area of a sector in polar coordinates. Those candidates who applied the correct formula were often able to write $\sin^2\theta$ in terms of $\cos 2\theta$ and carried out the subsequent integration correctly to find the area bounded by C_1 and the line segment OP. A significant number of candidates, however, could not find a correct method to integrate $\tan^2\theta$ The most successful method was to use the identity $\tan^2\theta = \sec^2\theta - 1$ although some other candidates found the correct area bounded by C_2 and the line segment OP by using integration by parts. Although a significant number of those candidates who obtained the correct values for the two areas went on to subtract these values to obtain the correct final answer, some others added their correct values and so failed to gain the final two marks.

Q8.

Part (a) was generally answered well with most candidates forming an equation in either $\sin\theta$ or r and solving it correctly to give the coordinates of P and Q. Candidates who tried to verify the coordinates of P generally failed to even verify them in both polar equations. Part (b) was the most demanding question on the paper. The majority of candidates scored at least 4 of the 11 marks by applying the formula $A = \frac{1}{2}\int r^2\,d\theta$ to find an area of a region partly bounded by the curve C and in doing so they correctly expanded $(1 - \sin\theta)^2$, wrote $\sin^2\theta$ in terms of $\cos 2\theta$ and integrated correctly. The errors occurred because many candidates did not use the correct limits. A significant number of candidates also failed to find, or even consider the need for, the area of triangle OPQ and so correct answers for the area of the shaded region R were generally only presented by the most able candidates.

Q9.

This question on polar coordinates proved to be the most demanding question on the paper. In part (a), the better candidates had no problems in finding the cartesian equation for C_1 and rearranging it by completing the squares so as to be able to deduce that the curve was a circle. Many other candidates failed to eliminate θ even after a page or more of working and abandoned part (a) of the question.

Many candidates scored heavily on the more familiar part (b)(i), finding the area bounded by the second curve C_2.

Proving that the two curves did not intersect in part (b)(ii) was a challenge, even for the better candidates, and it was rare to award all four marks for this part of the question. However, some excellent solutions were seen which used a variety of methods including forming and solving quadratic equations in either $\sin x$, $\cos x$, $\tan x$, $\sec x$, use of calculus and use of the R, α form.

In the final part of the question, those who used part (a)(ii) to find the area enclosed by C_1

and used it with their answer to part (b)(i) had no problems scoring both marks. Those who tried to use integration to find the area enclosed by C_1 did not analyse the problem fully and failed to score. A common misinterpretation of the word 'intersect' is indicated by the not uncommon answer 'Since C_1 and C_2 do not intersect the required area is just 16.5 π, the area bounded by C_2.'

Q10.

This question, which tested the area enclosed by a curve whose equation was given in polar form, was relatively poorly answered. Although almost all candidates obtained the correct integral for the area of the loop, a high proportion of these candidates could not then correctly integrate $(2 + \cos\theta)^2 \sin\theta$ with respect to θ. Only a minority of candidates realised that the integrand was the same as the derivative of $-\frac{1}{3}(2 + \cos\theta)^3$.

Most other candidates attempted to solve the integral by first writing the integrand as $2\sin\theta + 2\sin 2\theta + \cos^2\theta\sin\theta$ but the majority of these candidates could not then integrate $\cos^2\theta\sin\theta$.

Q11.

Some candidates integrated r instead of r^2 to find the area of the shaded region. If they had first stated the formula $A = \frac{1}{2}\int r^2\,d\theta$, which is given in the formulae booklet, before substituting for r further credit could have been awarded. The two most common errors were incorrectly squaring $6e^{-\frac{\theta}{\pi}}$ to get $6e^{-\frac{2\theta}{\pi}}$ and integrating $12e^{-\frac{\theta}{\pi}}$ incorrectly to get $-\frac{12}{\pi}e^{-\frac{\theta}{\pi}}$. The quality of sketches varied significantly with some even starting at the pole. A significant minority of candidates either just gave the coordinates of one end point, missing (1, 0) or gave a decimal approximation for e^2 or gave the incorrect answer $(e^2, 0)$. It was surprising to find some candidates giving the coordinates in reverse order. In part (c), most candidates formed a correct equation by equating the r terms but in general only the better candidates were then able to form and solve the resulting quadratic equation in $e^{\frac{\theta}{\pi}}$. Only a small number of candidates failed to reject the negative value for $e^{\frac{\theta}{\pi}}$ before going on to get the correct coordinates for the point P. Again candidates should use exact forms and not give a decimal approximation in place of $\pi \ln 3$. A common mistake seen in solving '$e^{\frac{\theta}{\pi}} = 1 + 6e^{-\frac{\theta}{\pi}}$' is illustrated by '$\frac{\theta}{\pi} = \ln 1 + 6\left(-\frac{\theta}{\pi}\right)$'.

Q12.

This question, which tested the areas of regions involving a curve whose equation was given in polar form, was relatively poorly answered. Full correct solutions were not often seen. In part (a), candidates generally wrote down the correct definite integral, then expanded $(1 + \tan\theta)^2$ and integrated $2\tan\theta$ correctly, but could not find a correct method to integrate $1 + \tan^2\theta$. Those who used the correct trigonometrical identity had no problem integrating the resulting $\sec^2\theta$ and completing the solution to reach the printed answer convincingly.

It was disappointing to find a significant minority of candidates not attempting part (b) having failed to obtain the printed answer in part (a). Most of the other candidates found the correct lengths for OP and OQ but some then wrote down an incorrect formula for the area of triangle OPQ. Some others lost the final mark because they did not give the area of the triangle in an exact form anywhere in their working despite the form of the printed answer in part (a).

Q13.

Parts (a) and (b) were generally answered well, but some candidates did not identify the critical value of 5 on their sketches. In part (c), although the usual errors were seen — for example the wrong expansion $(5 + 2\cos\theta)^2 = 25 + 10\cos\theta + 4\cos^2\theta$ or a sign error in the identity for $4\cos^2\theta$ in terms of $\cos 2\theta$ — most candidates had a thorough understanding of the method required to find the area of the region bounded by the curve. Although only a minority of candidates scored full marks for the final part of this last question, it is pleasing to report that a significant number of other candidates were awarded partial credit for finding an expression for OQ, although many lost at least one mark because they used $\pi - \alpha$ instead of $\alpha - \pi$ for θ. The most common wrong method seen involved the integral $\int_{-\pi+\alpha}^{\pi} \frac{1}{2} r^2 \, d\theta$, for which no credit was awarded.

Q14.

Most candidates showed that they understood how to find the area of the region but some lost accuracy marks by writing the wrong expansion for $(6 + 4\cos\theta)^2$ or making sign errors in the subsequent integration. Candidates found parts (b) and (c) much more of a challenge. In part (b), although many gave the correct values of r at P and Q, many then assumed that angle PQO was 90º to find the length of PQ, which they then used to 'prove' that angle PQO was 90º. Although some excellent solutions were seen for part (c), based either on the cosine rule for triangle POS or on the coordinates of P and S, the majority of candidates failed to present a full method or incorrectly assumed that PS was horizontal.

Q15.

A very high proportion of candidates showed knowledge of the two trigonometric identities required in part (a). Part (b) was also answered correctly by many candidates although the errors '$x^2 + y^2 = r$' and '$(r^2)^3 = r^5$' were not uncommon. Most candidates scored at least two of the three marks in part (c)(i), with some losing the final mark for either the wrong value $\frac{\pi}{4}$ or for giving their second value outside of the stated domain.

In the final part of the question, it was pleasing to find most candidates stating the general formula for the area of a sector in polar coordinates (as given in the AQA formulae booklet) and then substituting and expanding for r^2 correctly. It was pleasing to see a large number of candidates write $\sin^2 2\theta$ correctly in terms of $\cos 4\theta$ although, for others, all the usual errors were presented, many of which led to the 'correct' value $\frac{3\pi}{4}$ for the area of a loop. Although many candidates correctly used their values obtained in part (c)(i) for the limits of integration, some used different values without any explanation or justification. A very small number of candidates did not show their method of integration in reaching the answer $\frac{3\pi}{4}$. Such candidates lost the same significant number of marks as those who, for example, used the wrong identity ‘ $\sin^2\theta = \frac{1}{2}(1-\sin 4\theta)$ ‘ to integrate $\frac{1}{2}(1+2\sin 2\theta+\sin^2\theta)$ and obtained the answer $\frac{3\pi}{4}$.

Q16.

Those candidates who used $y = r\sin\theta$ and $x^2 + y^2 = r^2$ had no problem reaching the printed answer in part (a). In part (b), although almost all candidates started with a correct integral for the area bounded by the curve, errors in the squaring of 2 sinθ +5 were not uncommon. The method for integrating $\sin^2\theta$ was well understood but sign errors in the subsequent integration were seen. In the final part of the question, many candidates applied a correct method to find the points of intersection P and Q but a significant number could not present a valid method to find the area of the quadrilateral.

Q17.

In part (a), some candidates started with $\frac{1}{2}\int_0^{2\pi} 6(1-\cos\theta)\,d\theta$ and, without seeing a previous general formula, $\frac{1}{2}\int r^2\,d\theta$, examiners were unable to give the method mark. Although many candidates presented a correct solution for the area of the region bounded by the curve, errors in the squaring of 6(1– cosθ) were quite common. The method for integrating $\cos^2\theta$ was well understood and there was a lower proportion of sign errors in the subsequent integration than in the January 2006 exam. Although part (b) proved to be more of a challenge there were better than expected responses to both parts. The most common error was the use of ‘r = 9’ rather than the correct value r = 3. Also in part (i) some candidates gave the polar coordinates of the two points in the wrong form; basically (x, y). The most successful approach for finding the length of AB was to use AB = 2$r$sinθ (= 2y) although other correct methods included use of the cosine rule. Some candidates incorrectly assumed that triangle AOB was right-angled at O.